

1) Total ΔT_{fuel}

$$E_{TOT} = E_{TH} + E_D + E_{SFP} + E_{GFP}$$

$$\alpha_{th} = 10 \times 10^{-6}$$

$$t = 300 \text{ day} \left(\frac{24 \text{ h}}{1 \text{ day}} \right) \left(\frac{60 \text{ min}}{1 \text{ h}} \right) \left(\frac{60 \text{ sec}}{1 \text{ min}} \right) = 2.592 \times 10^6 \text{ sec}$$

$$E_{th} = \alpha \Delta T = (10 \times 10^{-6}) (1500\text{K} - 300\text{K})$$

$$= 0.012$$

$$E_D = \Delta \phi_0 \left(e^{\frac{B \ln 0.01}{C_D B_D}} - 1 \right)$$

$$B = \frac{\dot{F} t}{N_U}$$

$$M_{UO_2} = M_{UO_2} \approx 270 \frac{\text{g}}{\text{mol}}$$

$$N_U = \frac{\rho N_A}{A} = \frac{10.97 \frac{\text{g}}{\text{cm}^3} (6.022 \times 10^{23} \text{ atom/mol})}{270 \text{ g/mol}} = 2.45 \times 10^{22} \text{ atoms/cm}^3$$

$$B = \frac{(8 \times 10^{13}) (2.592 \times 10^6 \text{ sec})}{2.45 \times 10^{22} \text{ atoms/cm}^3}$$

$$= 0.0846 \text{ FIMA}$$

$$B_D = 5 \text{ MW/D} / \text{kgU} \div 950 = 0.005$$

$$B > B_D \text{ so max } E_D = \Delta \phi_0 = 0.015$$

(cont.)

$$E_{SFP} = 5.577 \times 10^{-2} \quad g\beta = 5.577 \times 10^{-2} \times 10.97 \times 0.0846 \quad \text{FIMA} \\ = 0.05176$$

$$E_{GFP} = 1.96 \times 10^{-28} \quad g\beta(2800-T)^{11.73} \times e^{-0.0162(2800-T)} \times \\ e^{-17.8 \frac{g\beta}{T}}$$

Where

$$g = 10.97 \quad \text{and} \quad \beta = 0.0846 \\ \text{and } T = 1500 \text{ K}$$

$$E_{GFP} = 6.69 \times 10^{-8}$$

$$E_{TOT} = 0.012 - 0.015 + 0.05176 + 6.69 \times 10^{-8} \\ = 0.0488 \times 100 = 4.88\%$$

2) total
Creep ϵ_{TA} $\leftarrow$ time

$\sigma = 60 \text{ MPa}$

$$\dot{\epsilon}_{\text{th}} = A_0 \left(\frac{\sigma}{G} \right)^n e^{(-Q/RT)}$$

$$\dot{\epsilon}_{\text{ir}} = C_0 \Phi^{c_1} \sigma^{c_2}$$

$$\Phi = 3 \times 10^{-11} \times \text{LFIR} = 7.5 \times 10^{-13}$$

$$\dot{\epsilon}_{\text{th}} = 3.14 \times 10^{24} \left(\frac{60 \text{ MPa}}{29208 \text{ MPa}} \right)^{250} e^{(-2.7 \times 10^5 / 8.314 \cdot 600 \text{ K})}$$

$$\dot{\epsilon}_{\text{th}} = 3.58 \times 10^{-13}$$

$$G = 4.2519 \times 10^{10} - 2.2185 \times 10^7 (600 \text{ K})$$

$$= 2.92 \times 10^{10} \text{ Pa} = 29208 \text{ MPa}$$

$$\dot{\epsilon}_{\text{ir}} = (1.654 \times 10^{-24}) (7.5 \times 10^{-13})^{0.85} (60 \text{ MPa})^1$$

$$\dot{\epsilon}_{\text{ir}} = 6.173 \times 10^{-11}$$

$$\dot{\epsilon}_{\text{tot}} = \dot{\epsilon}_{\text{ir}} + \dot{\epsilon}_{\text{th}} = 6.21 \times 10^{-11}$$

$$\text{Total creep} = (6.21 \times 10^{-11}) (t) = 0.00107$$

$$\text{time} = 200 \text{ day} \left(\frac{24}{1 \text{ day}} \right) \left(\frac{3600}{1 \text{ hr}} \right)$$

3)

2 pellets $t = 250$ days

5/8

$$T = 1400K$$

$$\dot{F} = 2 \times 10^{13} \text{ f/cm}^3 \text{ s}$$

$$a = \text{grain size} = 5 \mu$$

$$a = \text{grain size } 40 \mu$$

fraction of FGR in each

$$r = \frac{\dot{F} t}{a^2} \quad \checkmark \quad \text{from memory. thought it was } \frac{\dot{F} t}{2a} \text{ on exam sheet}$$

Assuming in-pile gas released
as heating after irradi. not described

$$D = D_1 + D_2 + D_3 = 2 \times 10^{13}$$

$$D_1 = 7.6 \times 10^{-6} \exp\left(\frac{-3.03}{8.6173 \times 10^{-5} \cdot 1400K}\right) = 9.4 \times 10^{-17} \quad \checkmark$$

$$D_2 = 1.41 \times 10^{-18} \exp\left(\frac{-1.19}{8.617 \times 10^{-5} \cdot 1400K}\right) \quad \checkmark$$

$$D_3 = (2 \times 10^{-30}) (2 \times 10^{13}) = 2 \times 10^{13} \quad \checkmark \quad \sqrt{2 \times 10^{13}} = 3.28 \times 10^6$$

$$D = 2 \times 10^{13}$$

got lost here...
looks like you didn't multiply

$$\tau = \frac{(2 \times 10^{13}) (250 \text{ days} \rightarrow \text{sec})}{(5 \times 10^{-6})^2} = 1.728 \times 10^{31} > \pi^{-2}$$

3)

$$\tau = \frac{2 \times 10^{13}}{2.5 \times 10^{-6} \text{ m}} = 8 \times 10^{18} > \pi^{-2}$$

$$\tau = \frac{2 \times 10^{13}}{2.4 \times 10^{-6} \text{ m}} = 8.3 \times 10^{18} > \pi^{-2}$$

$$\tau = \frac{(2 \times 10^{13}) (250 \text{ days} \rightarrow \text{sec})}{(40 \times 10^{-6})^2}$$

$$\pi^{-2} = 0.1013$$

$$\tau = 2.16 \times 10^{29} > \pi^{-2}$$

so choosing upper f for both

$$f = 1 - \frac{0.662}{\frac{Dt}{a^2}}$$

$$(1 - 0.92 \exp(-\pi^2 \frac{Dt}{a^2}))$$

$$f = 1 - 0.662$$

plugged for both in

$$D = 2 \times 10^{13}$$

$$t = 21600000 \text{ sec}$$

other process is correct
- but the 30 orders of magnitude difference should have a flag!

either = 5m or 40m

case with ~~than~~ $a = 5 \text{ m}$: $f = 1 - 3.83 \times 10^{-32}$

case with $a = 40 \text{ m}$: $f = 1 - 3.06 \times 10^{-30}$

Exam 3

Crystal Bowers

4) Typical charge state of U is +4 in UO_2 .

4/4 It can change to keep electrostatic neutrality. So when ~~insoluble~~ ^{soluble} fission products are introduced or metallic precipitate fission products (typically M^{3+}).

$$k_B = 8.6173 \times 10^{-5} \text{ eV/K}$$

$$R = 8.314 \text{ J/Kmol}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

4:45 PM

4 PM

G is in Pa!!

make sure
my σ are
in MPa

5/5 5) Oxygen concentration changes across the fuel pellet as fission events will produce more oxygen ~~and~~ and it will react with fission products. It can also ~~be~~ interact with cladding leading to its depletion in that area. Mo acts as a cap/buffer because Mo wants to form a metallic precipitate but when there's a high O potential, it can form an oxide.

3/3 6) Three properties that vary as a function of stoichiometry in UO_2 are melting temperature, thermal conductivity, and creep.

10/10 7) 5 Fission Products:

1. Soluble (γ): in cation sublattice causes swelling and decreases thermal conductivity (k).

2. Insoluble (Sr): in fluorite structure and ~~also~~ contributes to swelling ~~forms precipitates~~

3. Metallic (Mo): Forms as metallic precipitates, slightly increases thermal conductivity and contributes to swelling.

4. Volatile (Cs): Gas in pellet interior, solid at its exterior. Can corrode cladding if escapes and contributes to swelling.

5. Noble (Kr, Xe): Can increase plenum pressure if escapes. Decreases thermal conductivity in intragranular bubbles/voids or intergranular bubbles

8) 3 stages of FGR:

1. ~~1. Fission~~ Fission creates fission gas atoms located inside the grain; some form intragranular bubbles. The rest travel to the grain boundary.
2. Gas bubbles nucleate, ~~interconnect~~ grow, and interconnect on the grain boundary.
3. Gas atoms travel through interconnected bubbles to the free volume.

9) Creep is the mechanism of plastic deformation over time where the

5/3 applied stress is less than the yield stress. ^{Noble-Herring} ~~Noble-Herring~~ creep occurs via ~~bulk~~ diffusion in the bulk and is influenced by temperature.

8/8 10) High burnup structure (HBS) is caused by initially early in irradiation, local power is higher on the periphery of the pellet due to self-shielding.

More burnup leads to ^{238}U capturing more neutrons leading to the generation of ^{239}Pu and more local power and burnup increase. The crystal

structure has instability and goes through polygonization in UO_2 . In UO_2 the grains subdivide and become densely porous. Performance

aspects include an increase

in thermal conductivity and its capacity to retain fission product gases.

11) Irradiation growth of hcp ~~the~~ Zr

$\frac{b}{b}$ due to point defects is
interstitials and vacancies cluster to
form interstitial loops and
vacancy loops, respectively.

Interstitial loops form on Zr's
prismatic planes ~~the~~ causing elongation
in a and contraction in c.

Vacancy loops on basal
planes lead to
further contraction
in c.

12) Creep impacts Zr cladding

Over time by causing it to

7/8 creep down on the fuel pellets

and this interaction with pellets

ultimately leads to a tensile

stress in the cladding. The pellet

expands outwards making the rod

contract but ultimately over

time once the entire

fuel rod has interacted,

there's a decrease

in length. When only

in the center of the fuel

rod has interaction between

cladding and pellet there's

lengthening in the rod.

Thermal creep and irradiation

enhanced creep are the

creep ~~mechanisms~~ in cladding.

Wanted a bit
more of the
stages...

12
12

13) 4 conditions that must be met for SCC are ^{susceptible} suitable material, Sufficient time, Sufficient stress, and corrosive environment. Corrosive environment develops from the presence of volatile fission products such as the ^{cyclic} reaction of free I reacting with Zr and O ~~in the~~ which is met in PCI as O is coming from excess oxygen (O) from the fuel. Zr is the cladding. Stress condition is met in PCI as fuel is expanding and swelling ~~putting~~ causing tensile stress in the cladding, while there's also radial stress in the fuel and the cladding is creeping down. Sufficient time is met as there are four stages ~~in~~ starting with the SCC conditions like corrosive environment forming and typical operating cycles are 18-24 months. Suitable material is met because cladding ~~made~~ made of Zr is susceptible to SCC, continued on next page

13 (cont.) and alloying elements can only lead to slight benefits. Commercial PWRs and BWRs use Zr as the cladding. So this condition is met in PC1. A zirconia layer also can form on cladding as part of PC1.

14) Iodine (I) plays a role in the corrosion of Zr cladding by being separated from CsI through radiolytic decomposition. Excess oxygen is available from UO_2 . The I will interact with Zr to be ~~ZrI~~ ZrI_4 and then when ZrI_4 travels up the thermal gradient, the excess O will interact with the Zr to form an oxide that will form on the cladding and then I is free again to continue the cycle. Because of this oxide layer on the cladding walls, I attacks at the crack tip.

